## Plasma Proteins

**Plasma Proteins**: Proteins found in blood plasma that play various essential roles in the body.

## Important Plasma Proteins

**Albumins**: A major plasma protein, contributing to osmotic pressure and transport of substances.

- **g/L**: Grams per liter, a unit of concentration.
- **g%**: Grams per 100 mL, another unit of concentration.
- **mol. wt.**: Molecular weight, the weight of one mole of a substance.

**Globulins**: A group of proteins in plasma, including antibodies and enzymes.

**Fibrinogen**: A plasma protein essential for blood coagulation.

- **Coagulation**: The process of blood clotting.

**A/G Ratio**: The albumin to globulin ratio, used to assess protein levels in the blood.

## Other Physiological Important Plasma Proteins

- **Prothrombin**: A protein involved in blood clotting.
- **Angiotensinogen**: A protein that helps regulate blood pressure.
- **Complement System**: A part of the immune system that enhances the ability to clear pathogens.
- **Kinin System**: A group of proteins involved in inflammation and blood pressure regulation.
- **Blood Clotting Factors**: Proteins that help control bleeding.
- **Fibrinolytic System**: A system that breaks down blood clots.
- **Lipoproteins**: Proteins that transport lipids (fats) in the blood.
- **Haptoglobin**: A protein that binds free hemoglobin.
- **Fetuin**: A glycoprotein that regulates mineralization and bone formation.
- **α-antitrypsin**: A protein that protects tissues from enzymes of inflammatory cells.
- **Proteins C and S**: Proteins involved in regulating blood coagulation.
- **Transport Proteins**: Proteins that carry various substances in the blood.

## Albumin

**Content**: The amount of albumin in the plasma.

**Major Plasma Protein**: Indicates that albumin is one of the most abundant proteins in plasma.

**Molecular Weight**: The weight of the albumin molecule.

## Globulin

**Molecular Weight**: The weight of the globulin molecule.

**Types**: Refers to the different categories of globulins (α, β, γ).

- **β-globulin (Prothrombin)**: A type of globulin important for blood clotting.
- **γ-globulin**: Globulins that function as antibodies.

## Fibrinogen

**Required for Coagulation**: Indicates its essential role in blood clot formation.

**Converts to Fibrin**: Fibrinogen changes into fibrin, the protein that forms the mesh of a blood clot.

## Functions of Plasma Proteins

**Colloidal Osmotic Pressure**: The pressure exerted by proteins in the plasma that helps keep fluid in the bloodstream.

- **mmHg**: Millimeters of mercury, a unit of pressure.

**Protective Function**: The ability of plasma proteins, particularly antibodies, to defend against pathogens.

- **Antibodies**: Proteins that recognize and neutralize foreign substances.

**Coagulation**: The process of blood clotting involving plasma proteins.

**Transport**: The role of plasma proteins in carrying substances like carbon dioxide, hormones, metals, and drugs in the blood.

**ESR (Erythrocyte Sedimentation Rate)**: A blood test that measures how quickly red blood cells settle at the bottom of a test tube; influenced by fibrinogen levels.

**Viscosity**: The thickness or resistance to flow of plasma, influenced by plasma proteins and red blood cells.

**Buffer Mechanism**: The ability of plasma proteins to help maintain acid-base balance in the body.

**Protein Reserve**: Plasma proteins can be broken down into amino acids for use in protein synthesis during periods of starvation.

## White Blood Cells (Leucocytes)

**White Blood Cells (Leucocytes)**: The cells in blood that are part of the immune system and help the body fight infections and other diseases.

## General Features

**Formed Elements**: Components that are suspended in the blood, including cells and cell fragments.

**Nucleus**: A membrane-bound structure in cells that contains genetic material.

**Normal Count**: The typical number of white blood cells in a given volume of blood.

- **Cells/cu mm**: Cells per cubic millimeter, a unit for measuring cell concentration in blood.

**Life Span**: The duration of time that white blood cells live in the bloodstream.

## Types of Leucocytes

**Granulocytes**: A category of white blood cells that contain granules in their cytoplasm.

**Agranulocytes**: A category of white blood cells that do not have granules.

## Granulocytes

**Neutrophils**

**Total WBCs**: The percentage of neutrophils in the overall white blood cell count.

**Multilobed Nucleus**: A nucleus that has multiple lobes, characteristic of neutrophils.

**Phagocytosis**: The process by which cells engulf and digest foreign particles or microorganisms.

**Acute Infections**: Short-term infections that develop rapidly, such as pneumonia or appendicitis.

**Granules**: Small particles within cells that contain enzymes and other substances.

**Myeloperoxidase**: An enzyme found in neutrophils that helps kill bacteria.

**Proteases**: Enzymes that break down proteins and can destroy microorganisms.

**Thromboxanes**: Molecules released by neutrophils that influence blood vessel function.

**Leukotrienes**: Compounds that increase blood vessel permeability and attract white blood cells.

**Neutropenia**: A condition characterized by a low number of neutrophils.

**Aplastic Anemia**: A disorder where the bone marrow does not produce enough blood cells.

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**Eosinophils**

**Bilobed Nucleus**: A nucleus with two lobes, characteristic of eosinophils.

**Acidophilic Stain**: A type of stain that eosinophils pick up, indicating their affinity for acidic dyes.

**Eosinophilia**: An increase in the number of eosinophils, often due to allergic reactions or parasitic infections.

**Eosinopenia**: A decrease in eosinophils, which can occur during acute infections or with certain treatments.

**Mild Phagocytosis**: A lesser degree of engulfing and digesting foreign particles.

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**Basophils**

**Coarse Blue Granules**: Large granules in basophils that stain blue.

**Anticoagulant**: A substance that prevents blood clotting, like heparin released by basophils.

**Hypersensitivity Reactions**: Immune responses that are exaggerated, such as anaphylactic shock.

**Polycythemia**: A condition characterized by an increased number of red blood cells and sometimes basophils.

**Chronic Myeloid Leukemia**: A type of cancer that affects the blood and bone marrow.

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## Agranulocytes

**Lymphocytes**

**Large Round Nucleus**: A nucleus that is prominent and round in lymphocytes.

**Cytoplasm**: The gel-like substance inside a cell where organelles are suspended.

**Chronic Infections**: Long-term infections that can persist in the body, such as tuberculosis.

**Immunity**: The ability of the body to resist infections and diseases.

**T & B Lymphocytes**: Types of lymphocytes involved in the immune response; T cells help destroy infected cells, while B cells produce antibodies.

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**Monocytes**

**Largest WBCs**: Monocytes are the biggest type of white blood cell.

**Kidney-Shaped Nucleus**: A nucleus that resembles a kidney, typical of monocytes.

**Phagocytic Function**: The role of monocytes in engulfing and digesting pathogens.

**Macrophages**: A form of monocyte that has left the bloodstream and entered tissues, where it continues to perform phagocytosis.

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## Functions of Leucocytes

**Phagocytosis**: The action of white blood cells engulfing and digesting bacteria and other foreign materials.

**Antiallergic Effect**: The ability of eosinophils to counteract allergic reactions by inhibiting histamine release.

**Antibody Formation**: The process by which lymphocytes produce antibodies that help fight infections.

**Heparin Production**: The release of heparin by basophils, which helps prevent clotting within blood vessels.

**Trephine Formation**: The role of leucocytes in facilitating tissue growth and repair.

**Histamine**: A compound released during allergic reactions that increases blood flow and inflammation.

## BLOOD GROUP SYSTEMS

**Blood Group Systems**: Classification of human blood based on specific antigens present on the surface of red blood cells (RBCs).

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## 1. ABO SYSTEM

**Antigens**: Substances that induce an immune response; in this case, proteins found on the surface of red blood cells.

**RBC Membrane**: The outer layer of red blood cells that contains various proteins, including antigens.

**Blood Groups**: Categories of blood based on the presence or absence of specific antigens.

**Agglutinogen**: Another term for antigen; a substance that causes agglutination.

**Agglutinin**: Antibodies in plasma that can bind to agglutinogens and cause clumping of red blood cells.

**Agglutination**: The clumping together of red blood cells in response to an antibody binding to an antigen.

**Karl Landsteiner's Law**: A principle stating the relationship between the presence of antigens and the absence of corresponding antibodies.

**Subtypes**: Variants within a particular antigen group; A1 and A2 are subtypes of the A antigen.

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## 2. RH SYSTEM

**Discovered**: The identification of the Rh blood group system was made by Karl Landsteiner and Alexander Weiner.

**Agglutination with Rabbit Serum**: The reaction of red blood cells with serum from rabbits to identify Rh status.

**Antigens**: In the Rh system, C, D, and E refer to specific proteins present on red blood cells, with D being the most clinically significant.

**Genotypes**: The genetic constitution of an individual concerning the Rh factor, where DD and Dd indicate Rh-positive and dd indicates Rh-negative.

**Rh Antibodies**: Antibodies that develop in Rh-negative individuals when they are exposed to Rh-positive blood.

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## 3. SIGNIFICANCE OF RH GROUPING

**Transfusion**: The process of transferring blood from one person to another, where matching blood types is critical to avoid reactions.

**Pregnancy**: The condition where the mother carries a developing fetus; Rh incompatibility can lead to complications if the mother is Rh-negative and the fetus is Rh-positive.

**Hemolysis**: The destruction of red blood cells, which can occur when antibodies attack fetal red blood cells.

**HDN (Hemolytic Disease of the Newborn)**: A condition where the mother's antibodies attack the baby's red blood cells, leading to various health issues.

**Jaundice**: A yellowing of the skin and eyes due to high levels of bilirubin, often resulting from hemolysis.

**Erythroblastosis Fetalis**: A severe form of HDN characterized by the presence of immature red blood cells in the fetus due to hemolysis.

**Hydrops Fetalis**: A serious condition in the fetus characterized by an accumulation of fluid in body tissues due to severe anemia.

**Hemolytic Anemia**: A condition where red blood cells are destroyed faster than they can be made, leading to anemia.

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## 4. PREVENTION OF RH COMPLICATIONS

**Anti-D Antibodies**: Antibodies given to prevent Rh incompatibility complications in subsequent pregnancies.

**Fetal Protection**: Measures taken to shield the fetus from maternal antibodies that could cause harm.

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## 5. MNS SYSTEM

**Medicolegal Cases**: Legal cases involving medical evidence, where blood group testing can be crucial.

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## 6. IMPORTANCE OF BLOOD GROUPS

**Transfusion Reactions**: Adverse responses that can occur during blood transfusions if the donor and recipient blood types are not compatible.

**Disputed Parentage**: Legal cases where the parentage of a child is questioned, and blood group testing can provide evidence.

**Research**: Studies that explore the connections between blood groups and susceptibility to certain diseases.

**Anthropology**: The study of human societies, cultures, and their development, where blood group data can contribute to understanding genetic diversity.

**Ethnology**: The comparative and analytical study of cultures, which may utilize blood group information.

**Organ Transplantation**: The surgical process of transferring organs from one individual to another, where matching blood types is essential to prevent rejection.

**HDN Prevention**: The importance of Rh typing in preventing hemolytic disease in newborns.

## CLOTTING TIME

**Clotting Time**: The duration it takes for blood to form a clot after exposure to air or injury.

**Coagulation Process**: The transformation of blood from a liquid to a gelatinous or semisolid state due to blood vessel injury.

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## COAGULATION

**Coagulation**: The physiological process that causes blood to change from a liquid to a solid state, forming a clot.

**Physicochemical Changes**: Alterations in the physical and chemical properties of blood during the coagulation process.

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## STAGES OF COAGULATION

**Prothrombin Activator**: A complex formed from activated clotting factors that converts prothrombin into thrombin.

**Prothrombin**: An inactive plasma protein that is converted into thrombin during coagulation.

**Thrombin**: An enzyme that converts fibrinogen into fibrin, playing a crucial role in blood coagulation.

**Fibrinogen**: A soluble plasma protein that is converted into fibrin to form a blood clot.

**Fibrin**: An insoluble protein that forms the structural basis of a blood clot.

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## PROCESS OF CLOTING

**Clotting Factors**: Proteins in blood plasma that are essential for coagulation, acting as proenzymes that get activated during the clotting process.

**Cascade Mechanism**: A series of enzymatic reactions where each step amplifies the previous one, leading to clot formation.

**Platelet Adhesion**: The process by which platelets stick to the exposed surfaces of damaged blood vessels.

**Haemostatic Plug**: A temporary blockage formed by the aggregation of platelets at the site of injury.

**Vasoconstrictors**: Substances released by platelets that cause blood vessels to narrow, reducing blood flow.

**Platelet Factor III**: A phospholipid released by platelets that aids in the coagulation process.

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## GENERATION OF PROTHROMBIN ACTIVATOR

**Activated Factor X**: A key enzyme in the coagulation cascade that leads to the formation of prothrombin activator.

**Intrinsic Pathway**: The coagulation pathway activated by damage to blood vessels, involving factors present in the blood.

**Extrinsic Pathway**: The coagulation pathway activated by tissue injury, involving factors released from damaged tissues.

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## INTRINSIC PATHWAY

**Thromboplastin (Factor III)**: A protein released from damaged tissues that initiates the intrinsic pathway of coagulation.

**Factor VII**: A clotting factor that, when activated, helps activate Factor X.

**Factor IXa**: An activated clotting factor that contributes to the formation of prothrombin activator.

## EXTRINSIC PATHWAY

**Tissue Thromboplastin**: A protein released from damaged tissues that initiates the extrinsic pathway of coagulation.

**Calcium Ions**: Essential minerals that play a crucial role in various steps of the coagulation process.

**Factor V**: A clotting factor that is involved in the conversion of prothrombin to thrombin.

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## FORMATION OF FIBRIN

**Fibrin Polymer**: A stable structure formed by cross-linked fibrin threads, creating the framework of the blood clot.

**Fibrin-Stabilizing Factor (Factor XIII)**: A clotting factor that cross-links fibrin threads, strengthening the blood clot.

**Thrombin**: The enzyme responsible for converting fibrinogen into fibrin and facilitating clot stabilization.

**Fibrin Monomer**: The initial form of fibrin, which is soluble before polymerization into an insoluble clot.

**Clot**: The solid mass formed from fibrin that traps blood cells and serum, stopping bleeding.

## IMMUNITY

**Immunity**: The body's ability to defend itself against harmful microorganisms and foreign substances
**Microorganisms**: Tiny living organisms, such as bacteria, viruses, fungi, and protozoa, that can cause disease
**Foreign Substances**: Any substances that are not naturally found in the body and can trigger an immune response

## INNATE IMMUNITY

**Innate Immunity**: The non-specific defense mechanisms present at birth that provide immediate protection against pathogens
**First Line of Defense**: The initial protective barriers against infection, such as skin and mucous membranes
**Phagocytes**: Immune cells that engulf and digest pathogens and debris
**Complement System**: A group of proteins in the blood that assists in the destruction of pathogens
**Natural Killer (NK) Cells**: A type of immune cell that targets and kills infected or abnormal cells
**Inflammation**: A localized immune response that helps isolate and eliminate pathogens, characterized by redness, heat, swelling, and pain

## ADAPTIVE IMMUNITY

**Adaptive Immunity**: The specific immune response developed after exposure to pathogens, providing targeted protection
**B-cells**: Lymphocytes that produce antibodies and play a key role in humoral immunity
**T-cells**: Lymphocytes that are involved in cell-mediated immunity, directly targeting infected or abnormal cells

## HUMORAL IMMUNITY

**Humoral Immunity**: The aspect of adaptive immunity mediated by antibodies produced by B-cells
**Antibodies**: Proteins produced by B-cells that specifically bind to and neutralize antigens
**Plasma Cells**: Activated B-cells that secrete large amounts of antibodies into the bloodstream
**Memory B-cells**: Long-lived B-cells that provide rapid and robust responses upon re-exposure to the same antigen

## CELL-MEDIATED IMMUNITY

**Cell-Mediated Immunity**: The immune response that involves T-cells directly attacking infected or abnormal cells
**Helper T-cells**: A type of T-cell that activates other immune cells, including B-cells and cytotoxic T-cells
**Cytotoxic T-cells**: T-cells that kill infected or cancerous cells
**Memory T-cells**: Long-lived T-cells that remember previous infections and respond quickly upon re-exposure

## ANTIGEN-PRESENTING CELLS (APCs)

**Antigen-Presenting Cells (APCs)**: Immune cells that process and present antigens to T-cells to initiate an immune response
**Dendritic Cells**: Specialized APCs that capture and present antigens to T-cells, activating adaptive immunity
**Macrophages**: Large phagocytic cells that engulf pathogens and present their antigens to T-cells

## CYTOKINES

**Cytokines**: Signaling molecules that regulate immune responses and communication between cells
**Interleukins**: A type of cytokine that helps regulate immune cell growth, differentiation, and activation
**Interferons**: Cytokines that provide antiviral defense and modulate the immune response
**Tumor Necrosis Factors (TNFs)**: Cytokines involved in systemic inflammation and immune regulation

## IMMUNOLOGICAL MEMORY

**Immunological Memory**: The ability of the immune system to remember past infections and mount a faster response upon re-exposure
**Memory Cells**: Long-lived immune cells that retain the ability to recognize specific antigens for quicker responses

## VACCINATION

**Vaccination**: The process of stimulating adaptive immunity by introducing antigens without causing disease
**Antigens**: Substances that trigger an immune response, often derived from pathogens

## IMMUNODEFICIENCY

**Immunodeficiency**: A condition characterized by a weakened immune system, leading to increased susceptibility to infections

## AUTOIMMUNITY

**Autoimmunity**: A condition where the immune system mistakenly attacks the body's own tissues

**Rheumatoid Arthritis**: An autoimmune disease that primarily affects joints, causing inflammation and pain

## HYPERSENSITIVITY

**Hypersensitivity**: An exaggerated immune response to an antigen, leading to tissue damage and disease
**Allergies**: Overreactions of the immune system to harmless substances, such as pollen or food
**Delayed-Type Hypersensitivity**: A slower immune response involving T-cells, often seen in conditions like contact dermatitis

## SALTS

**Extrinsic Pathway**: The coagulation pathway initiated by tissue thromboplastin released from damaged tissues
**Prothrombin Activator**: A complex that catalyzes the conversion of prothrombin to thrombin
**Tissue Thromboplastin**: A protein released from damaged tissues that activates the extrinsic pathway

## CHELATING AGENTS

**Chelating Agents**: Chemicals that bind to metal ions, preventing them from participating in biochemical reactions
**Citrates**: Salts of citric acid that chelate calcium, preventing clot formation
**Oxalates**: Salts of oxalic acid that precipitate calcium, inhibiting coagulation
**Fluorides**: Compounds containing fluorine that interfere with calcium-dependent clotting mechanisms
**EDTA (Ethylenediaminetetraacetic Acid)**: A chelating agent that binds calcium, preventing blood clotting

## HEPARIN

**Heparin**: A natural anticoagulant that prevents clot formation by inhibiting thrombin and other clotting factors
**Mast Cells**: Immune cells that release histamine and heparin
**Basophils**: A type of white blood cell involved in inflammatory and allergic responses
**Sulphated Polysaccharide**: A complex sugar molecule containing sulfate groups that contribute to heparin's negative charge
**Antithrombin Factor III**: A protein that enhances heparin's ability to inhibit thrombin
**Thrombin**: An enzyme that converts fibrinogen into fibrin, leading to clot formation
**Heparin Cofactor**: A substance that enhances heparin's anticoagulant activity

**Protamine Sulphate**: A positively charged polypeptide that neutralizes heparin's anticoagulant effect

## DICOUMAROL

**Dicoumarol**: A Vitamin K antagonist that prevents clotting by inhibiting clotting factor synthesis
**Vitamin K Antagonist**: A substance that blocks the action of Vitamin K, which is essential for blood clotting
**Prothrombin**: A plasma protein that is converted into thrombin during coagulation
**Factors VII, IX, and X**: Clotting factors required for the blood coagulation process
**In Vivo**: A term meaning "within the living body"

## ERYTHROPOIESIS

**Precursor Stem Cells**: Undifferentiated cells that give rise to mature red blood cells

## IMPORTANT SITES OF ERYTHROPOIESIS

**Mesoderm**: The middle layer of embryonic tissue that forms blood and other structures
**Yolk Sac**: A membranous sac in early fetal life responsible for initial blood formation
**Extramedullary Haemopoiesis**: Blood cell production outside the bone marrow, occurring in organs like the liver and spleen under abnormal conditions
**Red Bone Marrow**: The active tissue in bones responsible for blood cell production
**Yellow Bone Marrow**: Fatty tissue in adult bones that replaces red marrow in long bone shafts

## STAGES OF ERYTHROPOIESIS

**Haemocytoblast**: A primitive, pluripotent stem cell capable of forming different blood cells
**Pluripotent Cell**: A cell that can develop into multiple types of blood cells
**Erythroid Series**: Cells that will develop into red blood cells
**Myeloid Series**: Cells that will develop into other blood cells, such as white blood cells and platelets
**Committed Stem Cells**: Cells destined to differentiate into a specific lineage

## PROERYTHROBLAST

**Precursor Cell**: An immature cell that will develop into a mature cell type
**Nucleus**: The central structure in a cell containing genetic material

**Nucleolus**: A dense region within the nucleus involved in ribosome production
**Proliferation**: Rapid cell division and growth

## EARLY NORMOBLAST

**Chromatin Threads**: Strands of DNA and proteins that condense during cell maturation
**Nucleoli**: Small structures inside the nucleus involved in ribosomal RNA production

## INTERMEDIATE NORMOBLAST

**Polychromatophilic**: A condition where the cytoplasm stains both acidic and basic dyes
**Haemoglobin Synthesis**: The process of producing haemoglobin, the oxygen-carrying protein in red blood cells

## LATE NORMOBLAST

**Pyknosis**: The condensation and shrinkage of the nucleus before its removal
**Extrusion**: The process of expelling the nucleus from the cell

## RETICULOCYTE

**Fine Reticulum**: A network of remnants of RNA seen in reticulocytes
**Nuclear Remnant**: The leftover RNA material from the nucleus
**Reticulocyte Count**: A measure of the percentage of immature red cells in circulation

## ERYTHROCYTE

**Peripheral Circulation**: The movement of blood cells through the bloodstream
**Morphology**: The structure and form of a cell

## BLOOD PRESSURE

**Lateral Pressure**: The force exerted by blood against vessel walls

## NORMAL BLOOD PRESSURE

**Systolic Pressure**: The pressure in arteries when the heart contracts
**Diastolic Pressure**: The pressure in arteries when the heart relaxes
**Pulse Pressure**: The difference between systolic and diastolic pressure
**Mean BP**: The average pressure in the arteries during a cardiac cycle

## REGULATION OF BLOOD PRESSURE

**Vasomotor Centre**: A brainstem region controlling blood vessel constriction and dilation
**Pressor Area**: A region in the vasomotor centre that increases BP
**Depressor Area**: A region in the vasomotor centre that decreases BP
**Sympathetic Tone**: The baseline activity of the sympathetic nervous system maintaining vessel constriction
**Thoracolumbar Outflow**: Nerve signals from the thoracic and lumbar spinal cord regulating vessel tone
**Higher Centres**: Brain areas (cerebral cortex, hypothalamus, limbic lobe) influencing BP

## SINOAORTIC REFLEX

**Baroreceptors**: Pressure-sensitive receptors detecting BP changes
**Carotid Sinus**: A dilated area in the carotid artery with baroreceptors
**Sinus Nerve (Glossopharyngeal Nerve)**: A nerve transmitting BP signals from the carotid sinus
**Aortic Arch**: A major artery from the heart containing baroreceptors
**Vagus Nerve**: A cranial nerve transmitting signals from the aortic arch
**Vasomotor Centre**: A brain region adjusting vessel tone
**Cardioinhibitory Centre**: A brain region reducing heart rate

## OTHER REFLEXES

**Left Ventricular Reflex**: A mechanism where increased pressure in the left ventricle lowers BP
**Pulmonary Reflex**: A response where lung congestion lowers heart rate and BP
**Coronary Chemoreflex**: A reflex where certain chemical injections in coronary arteries reduce BP
**Veratridine**: A plant-derived compound affecting BP
**Cushing's Reflex**: A response to high brain pressure causing increased BP and slow heart rate
**Bradycardia**: A slower-than-normal heart rate
**Higher Centre Reflex**: A response where emotions activate the sympathetic system to increase BP

## HORMONAL REGULATION

**Catecholamines**: Hormones like adrenaline and noradrenaline that increase BP
**Serotonin**: A neurotransmitter involved in vasoconstriction
**Angiotensin II**: A hormone that constricts blood vessels to raise BP

## LONG-TERM REGULATION OF BLOOD PRESSURE

**Fluid Balance**: The maintenance of water levels in the body
**Electrolyte Regulation**: Control of minerals like sodium and potassium in the blood

## RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM (RAAS)

**Renin**: An enzyme secreted by the kidneys when blood pressure drops
**Angiotensinogen**: A protein produced by the liver, converted into angiotensin I by renin
**Angiotensin I**: An inactive precursor that gets converted into angiotensin II
**Angiotensin-Converting Enzyme (ACE)**: An enzyme in the lungs that converts angiotensin I to angiotensin II
**Angiotensin II**: A hormone that constricts blood vessels and raises blood pressure
**Aldosterone**: A hormone from the adrenal glands that promotes sodium and water retention in the kidneys

## ANTIDIURETIC HORMONE (ADH)

**Pituitary Gland**: A small endocrine gland in the brain that releases ADH
**Osmolarity**: The concentration of solutes in body fluids
**Water Reabsorption**: The process of water being taken back into the bloodstream from the kidneys

## BARORECEPTOR REFLEX

**Baroreceptors**: Pressure-sensitive nerve endings in the carotid sinus and aortic arch
**Carotid Sinus**: A widened part of the carotid artery detecting blood pressure changes
**Aortic Arch**: A major artery carrying blood from the heart, containing baroreceptors
**Brainstem**: The lower part of the brain that regulates autonomic functions like blood pressure
**Heart Rate**: The number of times the heart beats per minute
**Blood Vessel Tone**: The degree of constriction or relaxation of blood vessels

## SYMPATHETIC NERVOUS SYSTEM

**Norepinephrine**: A neurotransmitter and hormone that increases heart rate and blood pressure
**Vasoconstriction**: The narrowing of blood vessels, leading to increased blood pressure

## ATRIAL NATRIURETIC PEPTIDE (ANP)

**Atria**: The upper chambers of the heart that release ANP in response to high blood volume
**Sodium Excretion**: The removal of sodium from the body through urine
**Vasodilation**: The widening of blood vessels, leading to lower blood pressure

## STABILIZATION OF BLOOD PRESSURE

**Vascular Resistance**: The force opposing blood flow within vessels

## HAEMOGLOBIN

**Red Pigment**: A colored substance that gives RBCs their red color
**Chromoprotein**: A protein combined with a pigment molecule
**Molecular Weight**: The sum of atomic weights of all atoms in a molecule

## NORMAL CONCENTRATION

**g% (Grams Percent)**: The amount of haemoglobin per 100 mL of blood

## FUNCTIONS

**Acid-Base Regulation**: Maintenance of pH balance in the blood
**Nitric Oxide (NO)**: A gas that relaxes blood vessels, causing vasodilation

## STRUCTURE OF HAEMOGLOBIN

**Subunit**: A structural part of a larger molecule
**Protoporphyrin IX**: A complex organic ring structure that binds iron
**$Fe^{2+}$ (Ferrous Iron)**: The form of iron that binds oxygen
**Polypeptide Chain**: A long chain of amino acids forming a protein

## TYPES OF HAEMOGLOBIN

**Thalassemia**: A genetic disorder causing abnormal haemoglobin production
**Sickle Cell Haemoglobin (Hb S)**: A mutated form of haemoglobin that makes RBCs sickle-shaped
**Osmotic Resistance**: The ability of cells to withstand changes in water movement

## FATE OF HAEMOGLOBIN

**Reticuloendothelial System**: A network of cells responsible for breaking down old RBCs
**Choleglobin**: A temporary haemoglobin breakdown product
**Biliverdin**: A green-colored pigment formed from haemoglobin breakdown
**Bilirubin**: A yellow pigment formed from biliverdin
**Glucuronic Acid**: A sugar acid that makes bilirubin water-soluble for excretion
**Bilinogens & Stercobilinogen**: Breakdown products of bilirubin in the intestines
**Urobilinogen**: A bilirubin breakdown product reabsorbed and excreted in urine

## ERYTHROBLASTOSIS FOETALIS

**Fetal Red Blood Cells**: Immature or mature red blood cells in the developing baby
**Destroyed**: Broken down or damaged beyond function
**Antibodies**: Proteins made by the immune system to fight foreign substances

## RH-INCOMPATIBLE PREGNANCY

**Rh Factor**: A protein found on red blood cells; people are Rh-positive if they have it and Rh-negative if they don't
**Rh-Incompatible**: When the mother is Rh-negative, and the fetus is Rh-positive

## IMMUNE RESPONSE

**Immune System**: The body's defense against infections and foreign substances
**Rh Antigen**: A protein present on Rh-positive red blood cells
**Foreign**: Not recognized as a natural part of the body
**Triggering**: Causing or starting a reaction

## PRODUCTION OF ANTIBODIES

**IgG Antibodies**: A type of immune protein that can cross the placenta to the fetus
**Placenta**: The organ that connects the mother and fetus, providing nutrients and oxygen

## HEMOLYSIS AND ITS EFFECTS

**Hemolysis**: The breakdown or destruction of red blood cells
**Anemia**: A condition where there are too few red blood cells to carry oxygen properly
**Erythroblasts**: Immature red blood cells produced in response to anemia

## SYMPTOMS

**Jaundice**: Yellowing of the skin and eyes due to bilirubin buildup
**Hepatosplenomegaly**: Enlargement of the liver and spleen
**Hydrops Fetalis**: Severe swelling (edema) in the fetus due to heart failure caused by severe anemia

## PREVENTION & TREATMENT

**Rh Immunoglobulin (RhIg)**: A medication that prevents the mother's immune system from attacking fetal red blood cells
**Blood Transfusion**: Replacing lost red blood cells with new ones
**Intrauterine Blood Transfusion**: A procedure to give blood to the fetus while still in the womb
**Early Delivery**: Inducing birth before the due date to treat complications